MIL-T-13918D

22 August 1973

SUPERSEDING
MIL-T-13918C
23 March 1967

MILITARY SPECIFICATION

TOOL KIT, VALVE SEAT RING INSERTER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of tool kit for removing and replacing valve seat ring inserts in cylinder blocks and heads of internal combustion engines.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATION

MILITARY

MIL-T-45542 - Tool Kits, Shop Sets, and Tool Sets (Both Common and Special), Preparation for Delivery of: General Specification For.

STANDARD

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5180

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LAWS AND REGULATIONS

29 CFR 1910 - OCCUPATIONAL SAFETY AND HEALTH ACT (OSHA) STANDARDS

(The Code of Federal Regulations (CFR) and Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards

Handbook H-28-Screwthread Standards

For Federal Service

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

Underwriters' Laboratories, Inc. (UL)

UL 45 - Standard For Electric Tools

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; Walt Whitman Road, Melville, NY 11746; or 1655 Scott Boulevard, Santa Clara, CA 95050.)

American Society for Testing and Materials (ASTM)

E 18 - Methods of Test for Rockwell Hardness and Superficial Hardness of Metallic Materials

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish a sample of the item to be procured under this specification for first article testing and approval (see 4.3 and 6.3).

3.2 Standard product. The tool kits delivered under this specification shall be the manufacturer's commercial product, except for any changes necessary to comply with specification requirements. All like items furnished on any one contract, including parts and subassemblies thereof, shall be new and interchangeable.

3.3 Codes and standards. The tool kits shall comply with the applicable requirements of UL 45 and OSHA Standards.

3.3.1 Compliance. Prior to approval of the first article, if one is submitted, or prior to commencing production, the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that the tool kits that he proposes to furnish under this specification meet the applicable requirements of UL 45 and OSHA Standards.

3.3.1.1 UL. Acceptable evidence of meeting the requirements of UL shall be the UL label, a UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating the tool kits have been tested and conform to UL Standard No. 45.

3.3.1.2 OSHA. The tool kits furnished under this specification shall comply with the applicable standards promulgated under the Occupational Safety and Health Act.

3.4 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial tool kit provided that the completed item complies with all provisions of this specification.

3.5 Design and construction. The tool kit shall consist of a carrying case and tools to replace or install valve seat ring inserts. The kit shall contain counterboring tools with adjustable cutters. The tool shall be used for installing ring inserts with outside diameters of 1-1/8 inch to 3-3/32 inch inclusive, in blocks or heads of internal combustion engine whose access to valve seats is not through the cylinders. In addition, they shall satisfactorily perform their operations without causing any damage to the block or head.

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3.5.1 Lubrication. All bearings shall be provided with oil cups or other suitable means for lubrication wherever required or bearings shall be sealed and lubricated. Gears shall be completely inclosed in a gear case packed with a standard gear grease for this type of application. All other moving parts shall be provided with means of being conveniently lubricated as may be required.

3.5.2 Threaded parts. Threads shall comply with Handbook H-28.

3.5.3 Knurls. Knurls shall be provided wherever gripping, assembling and disassembling or adjustments are normally performed with the hand.

3.5.4 Castings and forgings. All castings or forgings, when used in construction of these tools, shall be free from blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclosures, cold shuts, mismatching and other defects that may affect the structural strength of the forgings involved. Strength and other essential physical and chemical properties of the castings or forgings shall be adequate to meet the performance requirements specified herein. In no event shall welding, brazing, soldering, peening or plugging be used on castings or forgings for reclaiming any parts of these tools without authorization of the procuring activity.

3.5.5 Fastening devices. All nuts, bolts, pins and similar parts shall be installed with adequate means for preventing them from loosening or becoming out of adjustment. All fastening devices subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently deformed.

3.5.6. Component parts. The kit component parts shall consist of a base mounting assembly, counterboring attachment, spinning or peening attachment, pilots, motor operated drive unit, and insert driver and remover tools (see table I).

TABLE I. Component parts

Components	Size
Base assembly	
Case, valve seat inserter	
Cutter blades	Range 1-1/8" - 1-3/8"
" "	Range 1-3/8" - 1-23/32"
" "	Range 1-23/32" - 2-1/16"
" "	Range 2-1/16" - 2-13/32"

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TABLE I. Component parts (cont'd)

Components	Size
Cutter blades (cont'd)	Range 2-13/32" - 2-3/4"
" "	Range 2-3/4" - 3-3/32"
Cutterhead assembly	1-1/8" - 1-3/8"
" "	1-3/8" - 2-1/16"
" "	2-1/16" - 2-3/4"
Spinner blade	1-1/8" - 1-3/8"
" "	1-3/8" - 1-23/32"
" "	1-23/32" - 2-1/16"
" "	2-1/16" - 2-13/32"
" "	2-13/32" - 2-3/4"
" "	2-3/4" - 3-3/32"
Drive unit	
Ring driver head	1-3/32"
" " "	1-11/32"
" " "	1-15/32"
" " "	1-19/32"
" " "	1-23/32"
" " "	1-27/32"
" " "	1-31/32"
" " "	2-3/32"
" " "	2-7/32"
" " "	2-11/32"
" " "	2-15/32"
" " "	2-19/32"
" " "	2-23/32"
" " "	2-27/32"
" " "	2-31/32"
" " "	3-3/32"
Valve seat inserter	
Insert removing tool	

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TABLE I. Component parts (cont'd)

Components	Size
Ring driver pilot	5/16"
" " "	11/32"
" " "	3/8"
" " "	7/16"
" " "	1/2"
" " "	9/16"
Valve seat pilot	1/4"
" " "	9/32"
" " "	5/16"
" " "	11/32"
" " "	3/8"
" " "	7/16"
" " "	1/2"
Ring driver body	
Pilot wrench	

3.5.6.1 Insert removing tool. The insert removing tool shall be forged from a nominal 5/8-inch diameter, high grade tool steel, with a maximum overall length of 10-inches plus or minus 1/2-inch. One end of the removing tool shall be curved and beveled to fit under the insert to be removed. The opposite end shall be the striking end. This tool shall be capable of prying the insert from the engine head or cylinder block without evidence of distortion, bending, or breakage of the engine component or the tool, when tested in accordance with 4.5.2.1.

3.5.6.2 Counterboring tool. The counterboring tool shall be motor operated and shall be capable of producing a 50 microinch or better finish when operating with feed at operating speed. The counterboring tool shall have expandable cutting heads which are capable of operation at any diameter within their range. The counterboring tool shall have an automatic hand feed and a limit depth stop which is adjustable to within plus or minus 0.005 inch of the thickness of the valve insert. When tested in accordance with 4.5.2.2, the tool shall be capable of counterboring the required recess circular to within 0.002 inch total indicator reading; concentric with the valve guide hole to within 0.002 inch total indicator reading; and to within 0.005 inch per inch maximum taper with the guide hole in the engine block or head.

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3.5.6.2.1 Cutters. The counterboring tool shall be provided with one or more high-speed steel cutter blades mounted in a cutting head. The cutter blades shall have a hardness within the range of Rockwell C61-65 (see 4.5.1). The cutting head shall be supported by the base assembly and guided by a pilot fitted to the valve guide hole in the engine block or head. Cutter blades and cutter heads shall be capable of producing counterbores within the size ranges specified in 3.5. The counterbore produced shall provide for a press fit of the applicable insert when tested in accordance with 4.5.2.2.

3.5.6.3 Drive unit. The drive unit shall consist of the motor, housing, and cord with all attachments required to hold the drive unit in alignment for operation and for attaching the drive shaft to the tool heads for operation of the unit. Speed of drive unit shall be so selected that removal, counterboring, and peening operations can be performed at operating speed.

3.5.6.3.1 Electrical characteristics. The drive unit shall be designed for operation on a 60-Hertz, 120-volt power supply and shall conform to the applicable requirements of UL 45. Operating speed of the counterboring tool shall be between 50 and 60 revolutions per minute.

3.5.6.4 Base assembly. The base assembly shall have a cast vertical base, designed to be fastened to the block or head by the use of studs or cap screws normally used in an engine block or head. An arm, adjustable in height and in horizontal and vertical planes, and capable of rotating, shall be attached to this base. The arm shall be designed to guide the counterboring and peening or spinning attachments.

3.5.6.5 Pilots. Pilots shall be provided for the purpose of guiding the counterboring and peening or spinning tool. Sizes shall be as shown in table I. The pilots shall be designed to align the tool with new or worn valve guide holes in the engine block or head. The portion of the pilot making contact with the valve guide hole shall not restrict freedom of movement, yet provide a fit to enable concentricity to be within the tolerance limits specified in 3.5.6.2. The pilot hardness shall be within the range of Rockwell C50-60 (see 4.5.1).

3.5.6.6 Valve seat insert driver. The valve seat insert driver shall consist of pilots, driving rod and a set of driving heads. A driving head shall be provided for each size insert specified in 3.5, with a diameter 1/32 to 1/16 inch smaller than that of the applicable insert. Pilots of the sizes shown in table I shall be provided for alignment of the inserts during installation. When tested in accordance with 4.5.2.3, the driving rod and head shall remain in alignment and show no evidence of distortion.

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3.5.6.7 Peening or spinning tool. A peening or spinning tool attachment, designed for insertion in the counterboring attachment, shall be furnished. The tool shall be designed to displace (peen or spin) the block or head metal concentrically over the outer chamfered edge of each insert within the applicable size range, retaining (clinching) it. The tool shall be made of tool steel and shall be within the hardness range of Rockwell C55-58 (see 4.5.1) when tested in accordance with 4.5.2.4.

3.6 Finish. All exposed metal and working surfaces shall be finished in accordance with the manufacturer's standard commercial practice, except that no component or accessory shall be cadmium plated.

3.7 Marking. A noncorrosive metal nameplate, securely attached to the base assembly unit, shall be permanently marked with the manufacturer's name or readily identifiable trademark. Each cutter, cutting head, driving head and pilot shall have their size or size range permanently marked in the most convenient location practicable.

3.8 Handbook of instructions. The supplier shall furnish a handbook with each kit, containing operating instructions, necessary drawings and illustrations, maintenance instructions, safety precautions and a manufacturer's parts list.

3.9 Carrying case. The tool kit shall be furnished with a carrying case. The case shall be fabricated from a minimum thickness of No. 20 gage (.0375) sheet metal. The case shall be provided with a hinged cover, a carrying handle, and a hasp or similar means to keep the cover closed. The carrying case shall be of adequate size and partitioned to carry the tool kit equipment and spare cutters. The carrying case shall be finished in accordance with the manufacturer's standard commercial practice.

3.10 Workmanship. Exposed surfaces, except cutters, shall not contain sharp edges, burrs or splinters. The cutter blades shall be sharpened and shall not have any nicks or burrs. All rivets shall be tight. Castings and forgings shall be free from blowholes and porosity. Components of the tool kit shall not be damaged or impaired in any way and shall show no evidence of rust or corrosion.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements

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specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier shall insure that all components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards unless excluded, amended, modified or qualified in this specification or applicable purchase document.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.3).
2. Quality conformance inspection (see 4.4).

4.3 First article. When a first article is required, the tool kit shall be tested as specified in 4.5.1 thru 4.5.2.4. Failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.4.1 Component and material inspection. The supplier shall insure that all components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards unless excluded, amended, modified or qualified in this specification or applicable purchase document.

4.4.2 Intermediate testing. Cutters, pilots and peening or spinning tool shall be tested for hardness in accordance with 4.5.1 to determine compliance with 3.5.6.2.1, 3.5.6.5 and 3.5.6.7. A lot shall consist of a all cutters, pilots and peening or spinning tools as applicable, from all tool kits submitted for inspection at one time. The sample unit shall be one cutter, pilot, and peening or spinning tool as applicable. The inspection level shall be S-3 with an acceptable quality level (AQL) of 2.5 defects, expressed in terms of defects per hundred units.

4.4.3 End item inspection. The inspection lot shall be all tool kits offered for inspection at one time.

4.4.3.1 Visual inspection. The tool kits shall be examined for the defects in table II. The inspection level shall be II with an AQL

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(acceptable quality level) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

TABLE II. Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Any component or accessory cadmium plated.	X	
	Evidence of corrosion or rust.	X	
Carrying case	Not provided or not according to requirements specified.	X	
Construction and workmanship	Component fractured, malformed or otherwise impaired.	X	
	Component missing.	X	
	Functional component which requires abnormal force to operate.	X	
	Improperly assembled.	X	
	Insecure fastening devices.	X	
	Burrs or sharp edges not removed (except cutting edges of cutter blades).	X	
	Not clean.		X
	Cutters not adjustable	X	
	Electrical characteristics not as specified.	X	
Castings and forgings	Evidence of blow-holes, porosity, hard spots, shrinkage defects, cracks	X	
	Evidence of scale, inclosures, cold shuts, or mismatching.	X	
	Evidence of unauthorized reclaiming of castings or forgings (see 3.4.4).	X	
Handbook of instructions	Omitted or not as specified.		X
Marking	Missing, incomplete, illegible, or not permanent.		X

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4.4.3.2 Dimensional examination. Examination shall be made of tool kits for compliance with dimensions specified herein. Any dimensions not within the specified requirements shall be a defect. The inspection level shall be S-2 with an AQL of 2.5 defects, expressed in terms of defects per hundred units.

4.4.3.3 End item testing. When a first article is not required, the first produced tool kit shall be examined and tested as specified in 4.5.2 thru 4.5.2.4. Each sample tool kit shall be tested as specified in 4.5.2 thru 4.5.2.4. Failure to pass a test shall be cause for rejection of the inspection lot. The inspection level shall be S-2 with an AQL of 2.5 defects, expressed as defects per hundred units.

4.4.3.4 Code and standard examination. Certificate of compliance, certified test reports, approval labels, or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with specification requirements shall be examined and validated.

4.4.4 Examination of preparation for delivery. Examination shall be made to determine compliance with preservation, packaging and packing requirements in accordance with MIL-T-45542.

4.5 Tests.

4.5.1 Hardness tests. Hardness tests shall be in accordance with ASTM E 18.

4.5.2 Performance tests. The test specimens for the following tests shall be engine blocks, heads, or test plates, all of a cast iron-nickel or molybdenum alloy.

4.5.2.1 Removing insert test. Place insert removing tool into valve port opening so that beveled point of remover is under the base of the valve seat insert and the back of the remover is resting against the opposite wall of the insert. Striking the opposite end of the remover with a hammer, in the manner of striking a chisel, the beveled end of the removing tool shall, without fracturing or causing damage to the engine head, block or test plate, release or break out the worn or damaged valve seat insert. If an engine block or head is not available, this test shall be performed last, using the newly inserted seat in the test plate. Any nonconformance to 3.5.6.1 shall constitute a failure.

4.5.2.2 Counterboring test. Recesses shall be counterbored to fit the various sizes of valve seat inserts in accordance with the requirements specified in 3.5.6.2 and 3.5.6.2.1. Any nonconformance or evidence of damage or wear shall constitute a failure.

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4.5.2.3 Insert installation test. An insert within the specified size range (see table I) shall be selected and installed as follows:

(a) Set the limit stop on the counterboring tool (see 3.5.6.2), and counterbore a recess until the operation is halted by this stop.

(b) In accordance with 3.5.6.6, the insert shall be installed, using the drive tool from the tool kit, such that the upper face of the insert is within plus or minus 0.005 inch of the face of the test specimen. Any nonconformance shall constitute a failure.

4.5.2.4 Peening or spinning tool test. This test shall consist of peening or spinning (clinching) the above insert in accordance with 3.5.6.7. Any nonconformance, fracture of the test specimen, or distortion of the insert shall constitute a failure.

5. PREPARATION FOR DELIVERY

5.1 Levels A, B, and C. Tool kits shall be preserved, packaged, packed and marked for each level of protection in accordance with the applicable requirements conforming to type I of MIL-T-45542 (see 6.2).

6. NOTES

6.1 Intended use. Tool kits covered by this specification are intended for use in maintenance depots for replacing or installing valve seat inserts in heads and blocks of internal combustion engines.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.1, 4.3 and 6.3).
(c) Selection of applicable levels of preservation, packaging and packing (see 5.1).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

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6.4 Supersession data. Requirements for the type I, sizes 1 and 2, and type II, size 1 contained in MIL-T-13918C have been deleted from this specification. The kit covered by this specification was formerly identified as type II, size 2.

Custodians:

Army - GL
Air Force - 84

User activities:

Army - ME, AT, CE
Navy - MC, SH

Preparing activity:

Army - GL

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DOCUMENT IDENTIFIER AND TITLE

TOOL KIT, VALVE SEAT RING INSERTER MIL-T-13918D

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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